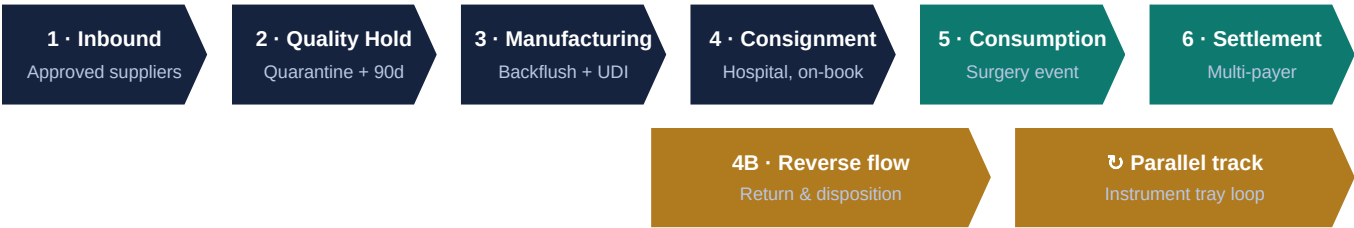


Orthopedic Implant Lifecycle

Full operational walkthrough — trigger, system action, database change, and handoff at every stage

The thread at a glance



Stage 4B (reverse flow) branches from Stage 4; reusable instrument trays run a parallel repeating loop rather than the one-way path.

Stage 1 — Inbound Procurement (Approved Suppliers)

Trigger	Purchase order confirmed against a vendor holding valid qualification status.
System action	Receipt operation type executes a push rule routing the incoming quant to <code>WH/Quarantine</code> (internal location type — remains on balance sheet).
Database change	<code>stock.lot</code> created with <code>x_udi_di</code> inherited from <code>product.template</code> ; <code>x_udi_pi</code> parsed from the inbound barcode scan into batch/serial/expiry segments via the MedTech extension layer (Object Schema §2.2).
Handoff	Purchasing → Quality (implicit, system-enforced — stock cannot bypass quarantine routing).

Stage 2 — Quality Isolation & 90-Day Hold

Trigger	Arrival in <code>WH/Quarantine</code> .
System action	Quality Control Point (native Quality app) auto-generates a Quality Check — visual inspection, storage-condition review against <code>x_storage_temp_min/max</code> (logger import or IoT feed, integration scoped in discovery), certificate-of-analysis verification.
Database change	<code>quality.check</code> record created, linked to <code>stock.lot</code> ; hold timer evaluated against <code>lot.create_date + 90 days</code> .
Release condition	Automated action requires both Quality Check = Pass and elapsed hold period before permitting a stock move out of quarantine — no manual override path in the standard flow. The hold is MDR Devices' own quality policy, enforced structurally; its exact regulatory/contractual driver is a named discovery confirmation.
Handoff	Quality → Logistics.

Stage 3 — Manufacturing Consumption & Finished UDI Generation

Trigger	Manufacturing Order scheduled against released raw material stock.
System action	Backflush consumption of raw material lot; finished goods lot created with a newly generated <code>x_udi_pi</code> (serial/batch/expiry computed at this stage) while <code>x_udi_di</code> is inherited from the finished product's template configuration.

Database change	New <code>stock.lot</code> record for finished good; <code>stock.move.line</code> records linking consumed raw lot(s) to the produced lot for full backward traceability.
Handoff	Manufacturing → Logistics (finished goods).

Stage 4 — Shipment to Hospital Consignment Hub

Trigger	Sale order confirmed (subject to the credential gate on <code>res.partner</code>). For planned surgeries, the delivery is scheduled against the confirmed surgery date — just-in-time replenishment driven by the clinical calendar, not generic lead times.
System action	Delivery order moves finished goods from <code>WH/Stock</code> to <code>WH/Consignment/{Hospital Name}</code> — an internal-type location, not a customer location. That location type preserves on-balance-sheet valuation while stock sits at the hospital (Object Schema §1.3). Design note: routing an SO delivery to an internal location is a deliberate custom-route configuration, not default SO behaviour — named in the Object Schema so it is designed and validated, not discovered at UAT.
Database change	<code>stock.quant</code> relocated within the internal location hierarchy; physical custody moves to the hospital without a valuation-triggering ownership change.
Handoff	Sales/Logistics → Field Service.

Stage 5 — Surgical Consumption

Trigger	Field Service activity marked “Surgery Complete,” referencing the consumed lot by UDI scan.
System action	Stock move from <code>WH/Consignment/{Hospital Name}</code> to a virtual <code>Consumed</code> location — this move is the designated revenue-recognition and invoicing trigger point, not the earlier delivery move (a deliberate departure from Odoo's default “invoice on delivery” pattern, configured and validated as such).
Database change	<code>stock.move</code> finalized; triggers the downstream settlement workflow.
Handoff	Field Service → Finance.

Stage 6 — Multi-Payer Financial Settlement

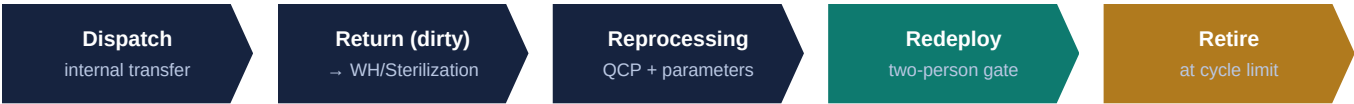
Trigger	Consumption move finalized.
System action	The extension layer's payer-split logic (Object Schema §4.2) generates the invoice set for the event — insurer and patient co-pay in the two-way case, additional tiers where the payer mix requires it. Stated precisely: a sale order natively invoices a single partner, so the split itself is scoped extension logic; the shared analytic distribution key (Cost Center / Payer Type / Product Line) that ties every resulting invoice back to the event is native.
Database change	Two (or more) <code>account.move</code> records created, cross-referenced via the shared analytic key to the originating consumption event; extension-layer rebate matching runs in parallel against the original purchase chain (Object Schema §4.3).
Handoff	Finance — closing the loop: full traceability from vendor PO to patient invoice in a single queryable thread.

Stage 4B — Reverse Flow: Consignment Return (Defective, Expired, or Cancelled Surgery)

Trigger	A consigned lot at <code>WH/Consignment/{Hospital Name}</code> is flagged by Logistics or Field Service as damaged, expired, or unused following a cancelled/postponed surgery — i.e., it will not reach Stage 5.
System action	Stock move from <code>WH/Consignment/{Hospital Name}</code> back to <code>WH/Quarantine/Returns</code> — a distinct internal location from the Stage 2 receiving quarantine, keeping inbound-new and returned-used dispositions from being confused in reporting.
Database change	Move recorded against the same <code>stock.lot</code> , preserving the full chain back to the original vendor lot — a return is a continuation of that lot's history, not a new record. The receiving quality reviewer sets <code>x_return_disposition</code> to one of four outcomes: Re-Quarantine (packaging intact, within sterility window, fresh Quality Check before re-entering <code>WH/Stock</code>), Scrap (write-off via <code>stock.scrap</code> referencing the original UDI), Return-to-Vendor (vendor-return delivery referencing the original PO for warranty/credit claims), or — for a return that doesn't scan cleanly — Under Investigation , routed to a manual match-and-flag queue for Quality to reconcile against shipment history. A companion <code>x_return_reason</code> field builds a queryable return-reason history.
Financial implication	If the original consignment placement had already triggered any billing event, disposition drives whether a credit note is required. If the lot was never consumed (Stage 5 never occurred), no financial reversal is needed — only the stock and traceability record change.
Handoff	Field Service/Logistics → Quality → Finance (conditional).

Parallel Track — Reusable Surgical Instrument Trays

Stages 1–6 and 4B describe single-use implant consumption. Instrument trays run a structurally different, repeating loop — see Object Schema §5 for field-level detail. In value-stream terms:



🔄 Passed, under-limit trays loop back to dispatch-ready stock.

Dispatch	Same mechanism as Stage 4 (internal-location transfer to the hospital).
Return	Instead of Stage 5's consumption move, a used tray moves to <code>WH/Sterilization</code> (internal location) — a return, not a consumption event; must not trigger revenue recognition or invoicing.
Reprocessing	A Quality Control Point gates the sterilization cycle, capturing autoclave parameters (temperature, duration, pressure) against the check, not just pass/fail; <code>x_sterilization_cycle_count</code> increments only on a passed cycle, and release requires the same two-step authentication gate as quarantine release (Object Schema §5.3) — a second reviewer countersigns before redeployment.
Redeploy / Retire	A passed, under-limit tray returns to dispatch-ready stock and re-enters this loop. A tray at or over <code>x_sterilization_cycle_limit</code> is blocked from further dispatch and routed to retirement — the same structural “no silent path around a control” pattern used throughout.

END-TO-END TRACEABILITY STATEMENT

At any point in this flow, a single query against `stock.lot` returns the complete chain: originating vendor, quality disposition, hold status, manufacturing consumption, consignment location history, consumption or return event, and resulting financial settlement — satisfying EU MDR Article 27 traceability and device-history documentation expectations (Device History Record under 21 CFR 820; the medical device file under ISO 13485). The majority of this chain runs on native Odoo Enterprise objects; UDI structuring, multi-payer & rebate settlement, and reusable-instrument cycle tracking are the deliberately scoped extensions, consistent with how comparable regulated-device engagements have closed these exact gaps previously.